

Molecular Property Prediction Based on Graph Contrastive Learning

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Abstract—Molecular Property Prediction (MPP) aids in drug discovery by predicting diverse molecular properties. This article presents a graph contrastive learning-based method for MPP. The approach effectively integrates substructure nodes into molecular graphs, enabling comprehensive molecular characterization. By combining specialized Graph Isomorphism Network (GIN) with contrastive learning, the method aims to improve prediction accuracy and generalization. The proposed method's effectiveness is confirmed by experimental results conducted on molecular property datasets.

Keywords—contrastive learning; molecular property prediction; deep learning

I. INTRODUCTION

The process of drug development is both intricate and protracted, necessitating substantial investments in manpower and research and development funds. MPP, which uses various experimental methods to predict the properties of molecules, is an important step in the discovery of new drugs. Accurately and reliably predicting molecular properties, including physical chemistry, biological activity, and ADME/T, in order to find drugs with ideal properties, is a long-term goal in the field of pharmacy.

Currently, a major challenge in the MPP field is the scarcity of labeled compound property data. With the increasing abundance of publicly available unlabeled molecular data, the rapid development of self-supervised learning (SSL) in the field of pharmaceutical chemistry has been facilitated. The current research introduces self-supervised pretraining techniques akin to those employed in CV and NLP. These methods initially pretrain the graph network on vast quantities of unlabeled molecular data, followed by fine-tuning on downstream labeled molecular data. Methods based on contrastive learning have recently become commonly employed as tasks for self-supervised pretraining of molecular representation learning models. The main principle of this method is to enable similar samples to learn representations that are close to each other, while representations of dissimilar samples are kept far apart from each other. Sun et al. [1] proposed a method that generates two enhanced views from local domain knowledge. Wang et al. [2] introduced three distinct graph enhancement techniques. Zhang et al. [3] decomposed the molecular graph into fragment pairs to create view representations for contrastive learning. Wu et al. [4]

proposed a novel dual-channel self-supervised pretraining strategy.

Molecules can be represented as graphs, where chemical bonds function as edges and atoms serve as nodes. Graph neural networks (GNNs) possess the ability to learn directly from such graphical representations, offering notable benefits in predicting molecular properties. Notably, the message passing neural network introduced by Gilmer et al. [5] has exhibited remarkable performance in quantum chemistry tasks involving graph-structured chemistry datasets.

Although graph neural networks are widely used in graph data analysis tasks and have achieved significant results, classical GNNs lack the ability to distinguish different graph structures, extract features insufficiently, and have poor generalization ability. Recently, researchers have conducted graph isomorphism tests on GIN, demonstrating that GIN can capture graph structures well[6]. Moreover, in chemical reactions, the substructures that directly participate in the reaction are often present within the molecule and play a decisive role in determining its properties. However, most existing methods use the 'atoms' and 'bonds' of molecules as the 'nodes' and 'edges' of GNNs, ignoring important substructure information in pharmaceutical chemistry. This makes it difficult to fully utilize the structural information of molecules when extracting feature information, resulting in poor accuracy in MPP.

This article introduces a MPP approach leveraging graph contrastive learning to tackle the aforementioned challenges, referred to as HCGIN. This method cleverly incorporates substructure nodes into molecular graphs, thereby achieving comprehensive and in-depth characterization of molecules. To effectively capture the intricate information encoded in the graph structure, we selected GIN as the backbone of our model. Furthermore, we employ contrastive learning methods as self-supervised tasks, aiming to further capture the structure and chemical properties of molecules by comparing the similarities and differences between different molecules.

The organization of the remainder of this article is outlined below. Section 2 offers a comprehensive explanation of the proposed MPP approach. In Section 3, we exhibited experimental outcomes across numerous MPP datasets to demonstrate the effectiveness of the method introduced in Section 2. Lastly, Section 4 offers a concise summary of the article's contributions.

II. METHODS

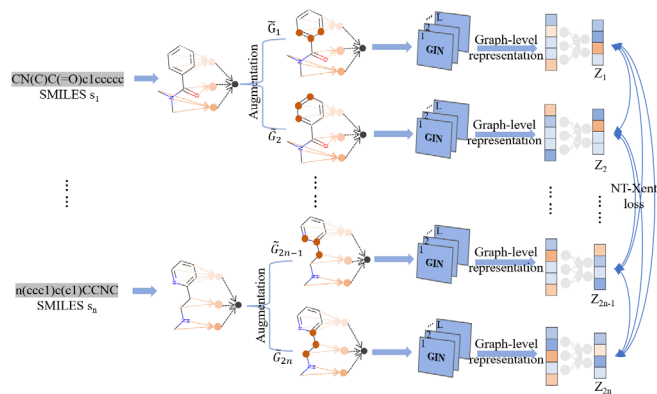


Figure 1. HCGIN framework

In this section, we provide a detailed explanation of the various components of the self-supervised learning framework HCGIN for molecular property prediction. The overall framework of HCGIN is shown in Fig.1.

A. Construction of Heterogeneous Molecular Graph

Molecules can be represented as 2D molecular diagrams, represented as $G = (V_a, E_a)$, where V_a and E_a represent nodes (atoms) and edges (chemical bonds), respectively.

Firstly, following the BRICS algorithm, the input molecular graph G is decomposed into a series of substructural fragments, where each fragment represents a subgraph of G . All obtained fragments are added to the molecular graph G as a single substructural node V_m , where (where k is the number of substructural fragments). Naturally, between each substructure node V_m and all the atomic nodes V_a it contains, a unidirectional edge E_{am} is added, connecting the atomic node V_a to the substructure node V_m .

Due to the need to obtain molecular representations through graph-level nodes, we add the graph-level node V_g within the G . The method involves linking all substructure nodes V_m to node V_g , and also forming a unidirectional edge E_{mg} from each substructure node V_m to the graph-level node V_g . The final heterogeneous molecular graph $\tilde{G}$ is represented as formula(1), which is visualized in Fig.2.

$$\tilde{G} = (\tilde{V}, \tilde{E}), \tilde{V} = [V_a, V_m, V_g], \tilde{E} = [E_a, E_{am}, E_{mg}] \quad (1)$$

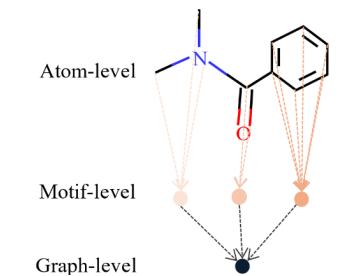


Figure 2. Heterogeneous molecular graph

B. Molecule Graph Augmentation

The fundamental concept of graph contrastive learning lies in creating positive and negative examples by augmenting graph structures. Positive examples are transformed structures from the same graph that remain similar, while negative examples are structures with notable differences from other graphs. The goal is to train a model where positive examples are close and negative examples are far apart in the projection space. Therefore, molecular graph augmentation methods are crucial for learning molecular representations.

On the basis of our heterogeneous molecular diagram $\tilde{G}$, we perform graph augmentation. For each heterogeneous molecular graph $\tilde{G}$, the subgraph deletion process starts from a randomly selected starting atomic node V_i . We mask the neighboring atoms of the initial atom V_i , proceeding to mask neighboring atoms until a designated proportion of the total atoms is masked. The way to mask the atomic node V_i is to assign a fixed value to its feature.

C. Graph Isomorphism Networks

Using the molecular graph augmentation strategy described in the previous section, the heterogeneous molecular graph $\tilde{G}$ is transformed into two distinct but related graphs: $\tilde{G}_i$ and $\tilde{G}_j$, where $i=2n-1$ and $j=2n$. We then separately input $\tilde{G}_i$ and $\tilde{G}_j$ into the GIN model, iteratively updating the node representations through neighborhood aggregation operations. The aggregation update rule for node features on the l^{th} layer of the GIN model is as follows:

$$h_v^l = \text{MLP}_a^l \left(h_v^{l-1} + \sum_{u \in \mathcal{N}(v)} (h_v^{l-1} + \text{MLP}_b^l(X_{uv}^b)) \right) \quad (2)$$

In the formula, h_v^l represents the feature representation of node v in the l^{th} layer, l represents the number of layers in the GIN. MLP_a^l and MLP_b^l represent the MLP for linear transformation of node and edge features at the l^{th} layer of the GIN, respectively. When $h_v^0 = X_v^a$, it represents the original input features of the node v , and X_{uv}^b represents the edge features between nodes u and v .

This model extracts the representation of graph-level nodes, $h_{v_g}^l$, and directly uses it as the molecular characterization of the molecule, eliminating the need for the READOUT operation in GIN.

D. Contrastive Learning

We use fully connected layers to encode molecular features $h_{v_g}^l$ and reduce dimensionality, resulting in the latent vector z_i for the molecule. This vector is utilized in graph contrastive learning, employing the NT-Xent loss function, which aims to maximize the distance between negative instances and minimize the distance between positive instances within the high-dimensional vector space. The NT-Xent loss function is represented as:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j) / \tau)}{\sum_{k=1}^{2N} 1_{\{k \neq i\}} \exp(\text{sim}(z_i, z_k) / \tau)} \quad (3)$$

Among them, l_{ij} represents the loss value calculated by the graph contrastive learning module, z_i and z_j are node representations of positive examples, z_k is the node representation of negative examples, N is the batch size, τ is a temperature parameter. The $\text{sim}(\cdot)$ function measures the similarity between two vectors. In our work, the cosine similarity function, shown in the following formula (4), is used.

$$\text{sim}(z_i, z_j) = \frac{z_i z_j}{\|z_i\|_2 \|z_j\|_2} \quad (4)$$

E. Molecular Property Prediction

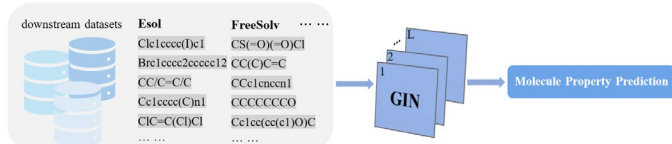


Figure 3. Molecular property prediction

The process of MPP is shown in Fig.3. The prediction model comprises a GIN backbone, identical in structure to the pre-trained model, and a fully connected layer. The prediction process is as follows:

$$y = f(h_g \cdot w_{fc} + b_{fc}) \quad (5)$$

h_g stands for the extracted feature representation of the graph-level node using the GIN backbone, w_{fc} denotes the weight matrix of the fully connected layer, b_{fc} signifies the bias of the fully connected layer, $f(\cdot)$ represents the ReLU activation function. y represents the predicted outcomes of the network model.

III. EXPERIMENTS

A. Datasets and Baseline Methods

For the pre-training of the model, we utilized the ZINC 250k dataset, which comprises approximately 250,000 drug-like molecules. Regression tasks can present unique challenges compared to classification tasks. Consequently, we conducted MPP experiments on five regression datasets from MoleculeNet, which collectively encompassed 16 regression tasks spanning various molecular properties across multiple fields.

To comprehensively assess the performance of HCGIN, we conducted a detailed comparison with various self-supervised learning SOTA as well as methods specifically designed for molecular graphs. These included GraphSAGE[7], ContextPred[8], GPT-GNN[9], InfoGraph[10], GraphLoG[11], GraphCL[12], JOAO[13], MolCLR[1], G-Motif[14], and HiMol[15].

B. Evaluation Criteria

The formula for the RMSE loss function is as follows. In the formula, t_i represents the true property value of i , y_i represents the MPP value of i , and n stands for the number of samples.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - t_i)^2} \quad (6)$$

The MSE loss function is presented in formula (7), where t_i , y_i , and n retain the same meanings as in the formula (6).

$$MAE = \frac{1}{n} \sum_{i=1}^n |h(x) - y| \quad (7)$$

C. Experimental Setups

We will randomly split the pre-training dataset into a training set and a validation set, with the former comprising 95% of the data and the latter accounting for the remaining 5%. For the MPP dataset, we use the scaffold segmentation method to create a training/validation/testing segmentation with a ratio of 80/10/10. In the pre-training stage, the model employs an Adam optimizer (with weight decay of 10^{-5}) to run for 100 epochs, optimizing the loss. And the learning rate is initially set to 5×10^{-4} , but after maintaining for 10 epochs, cosine learning rate decay is applied. The batch size is set to 256, and the hidden dimension in the model is set to 512. We use a 5-layer GIN as the backbone of the model, and its dropout rate is set to 0.3. In the finetuning stage, we perform 5 fine-tuning runs of the pre-trained model, each with 100 epochs. We present the mean test results from five separate runs. The drop rate is adjusted within [0.3, 0.7], and the batch size is adjusted within [32, 128].

TABLE I. MPP PERFORMANCE ON REGRESSION BENCHMARKS

Task Type	Regression(↓)				
Datasets	ESOL	FreeSolv	Lipo	QM7	QM8
Metrics	RMSE	RMSE	MAE	MAE	MAE
Molecules (Tasks)	1128 (1)	643 (1)	4200 (1)	7165 (1)	21786 (12)
GraphSAGE	2.575	5.051	1.212	164.062	0.0388
JOAO	1.355	4.280	0.771	270.839	0.0206
GPT-GNN	1.612	5.284	0.820	229.053	0.0204
MolCLR	1.333	3.285	0.720	104.184	0.0187
ContextPred	1.430	8.616	0.838	243.551	0.0205
InfoGraph	1.380	31.118	0.926	292.601	0.0192
GraphLoG	1.425	3.233	0.998	198.215	0.0903
GraphCL	1.265	5.569	0.782	285.967	0.0199
G-Motif	1.286	4.432	0.779	222.957	0.0203
HiMol	<u>0.833</u>	<u>2.283</u>	<u>0.708</u>	<u>91.501</u>	0.0199
HCGIN	0.824	2.049	0.694	70.054	<u>0.0191</u>

D. Results and Analysis

In the MPP regression dataset of Table 1, our model demonstrated excellent performance, outperforming most baseline models, which we attribute to its ability to capture higher-order structures. By utilizing the classic t-SNE method to intuitively explore the representation of nodes. We employ this method to visualize the molecular representations in the Lipo dataset, mapping them to a two-dimensional space. As shown in Fig.4, from the bottom right corner to the top left corner, the color of the dots gradually changes from blue to red, and from dark to light, indicating an increase in the octanol/water distribution coefficient of the drug. These

evidences the ability of our model to discern similarity disparities among molecules.

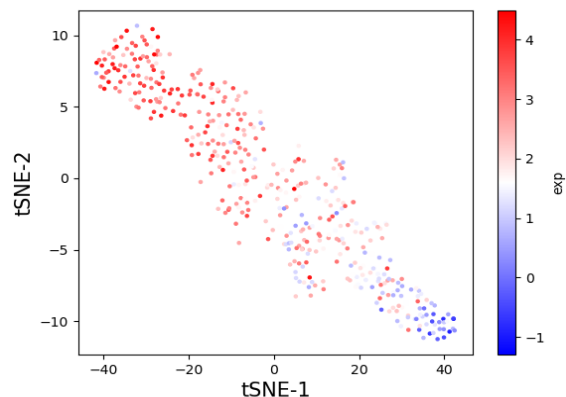


Figure 4. Visualization of molecular representations

To further investigate whether the model successfully captured substructure information that has a critical impact on drug properties, we fed all molecules from the pretrained molecular dataset into the trained model to obtain their high-dimensional feature representations. Subsequently, we employed the t-SNE method to visualize these high-dimensional vectors. Given the diverse range of substructures, we chose to display the top 15 substructures based on their frequency in the dataset. As depicted in the Fig.5, the points within each substructure cluster are tightly clustered, while those among different substructure clusters are relatively distant, exhibiting clear boundaries. This observation strongly indicates that the model has effectively captured the inherent structure and characteristics of diverse substructure fragments.

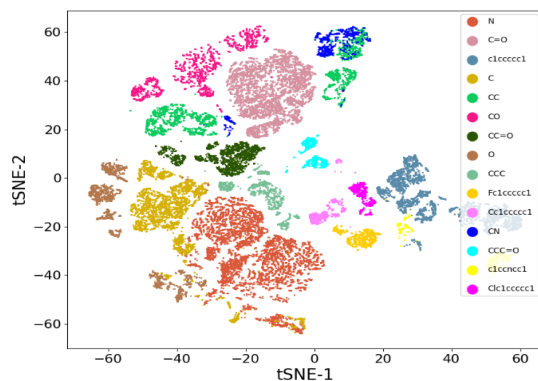


Figure 5. Visualization of substructure representations

IV. CONCLUSIONS

In this paper, we introduce a MPP approach leveraging graph contrastive learning. This method effectively incorporates the substructure of molecules on the basis of molecular diagrams, thereby achieving comprehensive and in-depth characterization of molecules. By using contrastive learning as a self-supervised task to compare the similarities and differences between different molecules, the structure and chemical properties of the molecules are further captured. The experimental results demonstrate that this approach can

enhance the precision of MPP tasks and demonstrate better performance than existing methods. Future research in this area could further explore the application of advanced neural network architectures and self-supervised learning strategies to optimize drug discovery processes and expand treatment options for patients.

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REFERENCES

- [1] Sun M, Xing J, Wang H, et al. MoCL: Contrastive Learning on Molecular Graphs with Multi-level Domain Knowledge[J]. In Proc. 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. pp. 3585–3594, 2021, <https://doi.org/10.48550/arXiv.2106.04509>
- [2] Wang Y, Wang J, Cao Z, et al. Molecular contrastive learning of representations via graph neural networks[J]. Nature Machine Intelligence, vol. 4, no.3, pp. 279–287, 2022, <https://doi.org/10.1038/s42256-022-00447-x>
- [3] Ziqiao Z, Ailin Xie, Jihong G, Shuigeng Z, et al. Molecular property prediction by semantic-invariant contrastive learning[J]. Bioinformatics, vol. 39, no.8, 2023, <https://doi.org/10.1093/bioinformatics/btad462>
- [4] Wu Y, Ni X, Wang Z, et al. Enhancing drug property prediction with dual-channel transfer learning based on molecular fragment[J]. BMC bioinformatics, vol. 24, no.1, 2023, <https://doi.org/10.1186/s12859-023-05413-x>
- [5] Gilmer J, Schoenholz S S, Riley P F, et al. Neural message passing for quantum chemistry[C]/International conference on machine learning. PMLR, pp. 1263–1272, 2017, <https://arxiv.org/abs/1704.01212>
- [6] Xu K, Hu W, Leskovec J, et al. How powerful are graph neural networks?[J]. International Conference on Learning Representations, 2019b. <https://doi.org/10.48550/arXiv.1810.00826>
- [7] Hamilton W, Ying Z, Leskovec J. Inductive representation learning on large graphs[J]. Advances in neural information processing systems, 2017, <https://arxiv.org/abs/1706.02216>
- [8] Hu W, Liu B, Gomes J, et al. Strategies for Pre-training Graph Neural Networks[C]/International Conference on Learning Representations, 2020, <https://doi.org/10.48550/arXiv.1905.12265>
- [9] Hu Z, Dong Y, Wang K, et al. Gpt-gnn: Generative pre-training of graph neural networks[C]/Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining. pp. 1857–1867, 2020, <https://doi.org/10.1145/3394486.3403237>
- [10] Sun F Y, Hoffmann J, Verma V, et al. InfoGraph: Unsupervised and Semi-supervised Graph-Level Representation Learning via Mutual Information Maximization[J]. 2019, <https://doi.org/10.48550/arXiv.1908.01000>
- [11] Xu M, Wang H, Ni B, et al. Self-supervised graph-level representation learning with local and global structure[C]/International Conference on Machine Learning. pp. 11548–11558, 2021, <https://arxiv.org/pdf/2106.04113>
- [12] You Y, Chen T, Sui Y, et al. Graph contrastive learning with augmentations[J]. Advances in neural information processing systems, pp. 5812–5823, 2020, <https://arxiv.org/abs/2010.13902>
- [13] You Y, Chen T, Shen Y, et al. Graph contrastive learning automated[C]/International Conference on Machine Learning. pp. 12121–12132, 2021, <https://arxiv.org/abs/2106.07594>
- [14] Rong Y, Bian Y, Xu T, et al. Self-supervised graph transformer on large-scale molecular data[J]. Advances in neural information processing systems, pp. 12559–12571, 2020, <https://arxiv.org/abs/2007.02835>
- [15] Zang X, Zhao X, Tang B. Hierarchical molecular graph self-supervised learning for property prediction[J]. Communications Chemistry, vol. 6, no.1, pp. 34, 2023, <https://doi.org/10.1038/s42004-023-00825-5>